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NumPy Cheat Sheet

Arrays, maths & numerical computing for data science.

CREATING ARRAYS

```
np.array([1,2,3])
np.zeros((3,4))
np.ones((2,3))
np.eye(3)
np.arange(0,10,2)
np.linspace(0,1,5) # 5 ev
np.random.rand(3,3)
```

ARRAY INSPECTION

```
a.shape
a.ndim
a.size
a.dtype
a.itemsize
a.T
a.flatten() # 1D copy
```

RESHAPING

```
a.reshape(3,4)
a.reshape(-1,1)
a.ravel()
np.expand_dims(a,axis=0)
np.squeeze(a)
np.stack([a,b])
np.concatenate([a,b],axis
```

INDEXING & SLICING

```
a[0]
a[-1]
a[1:3]
a[::2]
a[1,2]
a[:,1]
a[a>5]
```

MATHS OPERATIONS

```
np.sum(a)
np.mean(a)
np.std(a)
np.min(a) / np.max(a)
np.median(a)
np.cumsum(a)
np.diff(a)
```

BROADCASTING

```
a + b
a * 2
a @ b
np.dot(a,b)


np.broadcast_to(a,(3,4))
```

LINEAR ALGEBRA

```
np.linalg.inv(a)
np.linalg.det(a)
np.linalg.eig(a)
np.linalg.svd(a)
np.linalg.norm(a)
np.linalg.solve(A,b)
np.linalg.matrix_rank(a)
```

SORTING & SEARCHING

```
np.sort(a)
np.argsort(a)
np.argmin(a)
np.argmax(a)
np.where(a>0,a,0)
np.unique(a)
np.searchsorted(a,v)
```

RANDOM

```
np.random.seed(42)
np.random.rand(3,3)
np.random.randn(3,3) # no
np.random.randint(0,10,5)
np.random.choice(arr,5)
np.random.shuffle(a) # in
np.random.permutation(a)
```

SET OPERATIONS

```
np.unique(a)
np.intersect1d(a,b)
np.union1d(a,b)
np.setdiff1d(a,b)
np.in1d(a,b)
np.isin(a,b)
np.setxor1d(a,b)
```

TYPE CASTING & SAVE

```
a.astype(np.float32)
a.astype(np.int64)
np.float16 / int32 / comp
np.save('file.npy',a)
np.load('file.npy')
np.savetxt('f.csv',a,deli
np.loadtxt('f.csv',delimi
```

USEFUL TRICKS

```
np.clip(a,0,1)
np.nan_to_num(a)
np.isnan(a)
np.percentile(a,75)
np.histogram(a,bins=10)
np.tile(a,(2,3))
np.pad(a,1,'constant')
```